Sequential Tradeability Testing for Alpha Signals

A Bayesian optimal stopping view of alpha validation

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MSE342 Project

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When should a we stop researching a signal and admit it to live trading?

Portfolio optimization needs trusted alpha

$$
\begin{aligned}
\max_{w} \quad & \alpha^\top w \;-\; \frac{\gamma}{2} w^\top \Sigma\, w \;-\; \mathrm{Cost}(w - w_0) \\
\text{s.t.} \quad & \mathbf{1}^\top w = 1, \qquad \ell \le w \le u, \\
& F^\top w = f, \qquad \|w - w_0\|_1 \le \tau.
\end{aligned}
$$

w weights, w_0 current holdings, $\Sigma = F\Sigma_f F^\top + D$ factor risk, $\mathrm{Cost}(\cdot)$ spread + impact, F factor loadings, τ turnover budget.

Risk, constraints, and trading costs can be modeled carefully. **But what admits α?**

The upstream question

α is not a given input: it is an uncertain object learned from noisy evidence, and the most fragile input in the pipeline. This project is the *alpha-validation layer*: which signals have enough evidence to be admitted as α. It is not a replacement for the optimizer.

The object of the project

research evidence arrives over time $\implies$ continue, activate, or reject

Thesis

Alpha validation is a stopping problem under posterior uncertainty: evidence acquisition has option value, while delay, decay, and implementation costs enter the payoff.

EKV supplies a tractable posterior diffusion. The project replaces squared-error terminal loss with an economic admission payoff.

Fixed horizons are not admission rules

Fixed-horizon test

$$\text{at } T: \quad \text{is the signal significant?}$$

- conditions on an exogenous date;
- separates statistical evidence from implementation value;
- delays both activation and rejection.

Sequential admission

$$\tau: \quad \text{continue, activate, or reject}$$

- endogenous decision time;
- stopping region partitions activation and rejection;
- continuation prices one more observation.

Storyline

1. EKV: learn an unknown drift from noisy Brownian evidence.
2. Key identity: posterior uncertainty is also the speed of learning.
3. Project move: replace EKV's terminal statistical loss with an economic alpha payoff.
4. Stress test: Gaussian beliefs capture learning option value but rule out a point-mass null.
5. Extension: a three-state prior makes the null-alpha state explicit.
6. Evidence: OSAP and WRDS experiments compare sequential admission with fixed horizons.

EKV starts with drift learning

$$Y_t = X\,t + W_t, \qquad t \geq 0.$$

- X is an unknown constant drift drawn from a prior μ.
- Y_t is cumulative noisy evidence.
- $\mathcal{F}_t^Y = \sigma(Y_s : 0 \leq s \leq t)$ is the information available to the decision maker.

Alpha interpretation

For signal validation, X is latent alpha strength (the signal's risk-adjusted excess-return edge) and Y_t is normalized cumulative return evidence.

Bayes' rule gives the posterior state

$$\mu_{t,y}(du) = \frac{\exp\{uy - u^2 t/2\}\mu(du)}{\int_{\mathbb{R}} \exp\{vy - v^2 t/2\}\mu(dv)}.$$

$$\widehat{X}_t = G(t, Y_t)$$

$$G(t,y) = \mathbb{E}[X \mid Y_t = y]$$

$$H(t,y) = \mathrm{Var}(X \mid Y_t = y)$$

$$\partial_y G(t,y) = H(t,y)$$

The posterior mean and posterior variance are the low-dimensional state variables.

The filtering identity

$$\widetilde{W}_t = Y_t - \int_0^t \widehat{X}_s \, ds \quad \text{(innovation, a } \mathcal{F}^Y\text{-Brownian motion)}$$

$$d\widehat{X}_t = \Psi(t, \widehat{X}_t)\, d\widetilde{W}_t, \qquad \Psi(t, \widehat{X}_t) = \mathrm{Var}(X \mid \mathcal{F}_t^Y).$$

Intuition

The posterior mean is an $\mathcal{F}_t^Y$-martingale. The single quantity Ψ is both the remaining estimation error and the local volatility of belief revisions: the state variable and the learning rate at once.

EKV's optimal stopping problem

$$\inf_{\tau} \mathbb{E}\left[(X - \widehat{X}_{\tau})^2 + c\tau\right].$$

EKV study the optimal time to stop observing and report the posterior mean.

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$$\mathbb{E}[\Psi(\tau, \widehat{X}_\tau)] = \mathrm{Var}(X) - \mathbb{E}\Big[\int_0^\tau \Psi^2(s, \widehat{X}_s)\, ds\Big].$$

(Posterior variance $=$ mean-squared error; Itô on the martingale $\widehat{X}$ gives quadratic variation Ψ^2.) Expected posterior variance decreases by accumulated squared learning speed.

The EKV problem becomes a running-cost problem

$$\inf_{\tau} \mathbb{E}\left[\int_0^{\tau} \left(c - \Psi^2(s, \widehat{X}_s)\right) ds\right].$$

Cost of waiting

$$c$$

observation and opportunity cost

Benefit of waiting

$$\Psi^2$$

instantaneous variance reduction

Two EKV benchmarks we use to validate the math

Gaussian prior

$$X \sim N(m_0, q_0)$$

$$q(t) = \frac{q_0}{1 + q_0 t}, \qquad dm_t = q(t)\, d\widetilde{W}_t.$$

$$\tau^\star = \left(\frac{1}{\sqrt{c}} - \frac{1}{q_0}\right)^+.$$

Symmetric Bernoulli prior

$$X \in \{-\beta, \beta\}, \qquad \Psi(x) = \beta^2 - x^2.$$

Continuation is a symmetric band in posterior mean: the continuous-time sequential probability ratio test geometry.

The project changes the terminal reward

EKV: estimate accurately $\Longrightarrow$ finance: admission payoff

Scope of transfer

EKV gives the posterior state process. The finance boundary is obtained by changing the reward functional and solving the induced obstacle problem.

Variable translation

EKV object	Alpha validation object
X	unknown alpha strength
Y_t	cumulative normalized evidence
$\widehat{X}_t$	posterior mean alpha
Ψ	posterior uncertainty and learning speed
c	research cost, delay, opportunity cost
τ	activation or rejection time

Economic activation payoff

$$mz - \frac{\lambda}{2}(\sigma_r^2 + q)z^2 - \kappa.$$

$m = \widehat{X}$ posterior mean alpha, $q = \Psi$ posterior parameter risk, z exposure, λ risk aversion, σ_r^2 return variance, κ implementation hurdle.

Economic activation payoff

$$mz - \frac{\lambda}{2}(\sigma_r^2 + q)z^2 - \kappa.$$

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The objective is concave in z; the optimum and optimized value are closed form:

$$z^\star = \frac{m}{\lambda(\sigma_r^2 + q)}, \qquad \Phi(m, q) = \left(\frac{m^2}{2\lambda(\sigma_r^2 + q)} - \kappa\right)^+.$$

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$$|m| > \sqrt{2\lambda(\sigma_r^2 + q)\kappa}$$

is the static activation threshold. The stopping boundary also incorporates the option value of future evidence.

The finite-horizon stopping problem

$$V(t,m) = \sup_{\tau\in[t,T]} \mathbb{E}_{t,m}\left[e^{-\rho\tau}\Phi(m_\tau, q(\tau)) - \int_t^\tau c\,ds\right].$$

The finite-horizon stopping problem

$$V(t,m) = \sup_{\tau \in [t,T]} \mathbb{E}_{t,m}\left[e^{-\rho\tau}\Phi(m_\tau, q(\tau)) - \int_t^\tau c\,ds\right].$$

With obstacle $g(t,m) = e^{-\rho t}\Phi(m, q(t))$ and generator $\mathcal{L}_t V = \frac{1}{2}q(t)^2 V_{mm}$, the value solves the variational inequality

$$\max\{\, g - V,\ V_t + \mathcal{L}_t V - c\} = 0, \qquad V \geq g.$$

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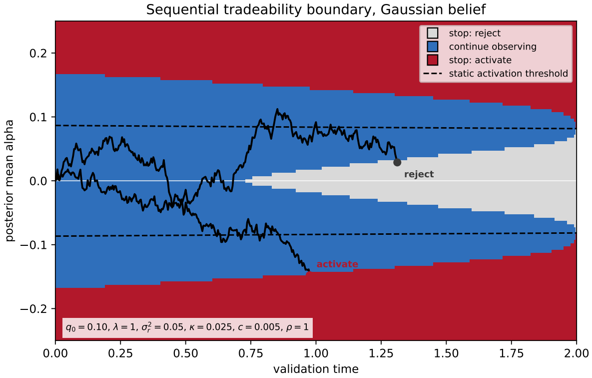
Continuation region $\mathcal{C} = \{V > g\}$, stopping region $\mathcal{S} = \{V = g\}$, separated by a free boundary with value matching $V = g$ (smooth pasting where regular).

Convention: discounting sits in the time-dependent obstacle $e^{-\rho t}\Phi$ (calendar-time value), so no $-\rho V$ term; equivalently $e^{-\rho(\tau - t)}$ would move it into the continuation operator.

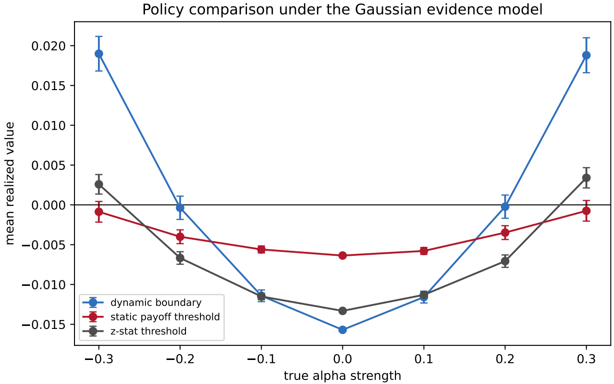
What the boundary means



- Red: stop and activate ($\mathcal{S}$).
- Gray: stop and reject ($\mathcal{S}$).
- Blue: continue observing ($\mathcal{C}$).
- Dashed: static activation threshold.
- Lines: two posterior-mean paths.

The continuation region $\mathcal{C}$ is a value-of-information region, not a positive-payoff region.

First stress test: Gaussian beliefs



Model success

For large $|\theta|$, dynamic stopping extracts learning option value.

Model limitation

At $\theta = 0$, the Gaussian prior activates about 90.1% of null paths.

The Gaussian prior rules out a point-mass null

$$X \sim N(0, q_0) \quad \Rightarrow \quad \mathbb{P}(X = 0) = 0.$$

A diffuse Gaussian prior treats the null as an event of posterior probability zero. That is a structural mismatch when the candidate set may contain signals with no admissible economic alpha.

Implication

Introduce an explicit null-alpha state and let posterior evidence move mass into or out of that state.

A three-state prior for alpha admission

$$\theta \in \{-a, 0, +a\}.$$

State	Interpretation
$-a$	negative alpha
0	null alpha
$+a$	positive alpha

The state space now separates sign uncertainty from the probability of no admissible alpha.

Posterior probabilities under the three-state prior

$$\pi_i(t, y) = \frac{p_i \exp\{\theta_i y - \theta_i^2 t/2\}}{\sum_j p_j \exp\{\theta_j y - \theta_j^2 t/2\}}.$$

$$m(t, y) = \sum_i \pi_i(t, y)\theta_i, \qquad q(t, y) = \sum_i \pi_i(t, y)\theta_i^2 - m(t, y)^2.$$

Near-zero cumulative evidence increases the posterior mass on $\theta = 0$, because nonzero drift states are less consistent with a persistent absence of direction.

False-discovery penalty

$$\Phi_\eta(t,y) = e^{-\rho t}\left(\frac{m(t,y)^2}{2\lambda(\sigma_r^2 + q(t,y))} - \kappa - \eta\pi_0(t,y)\right)^+ .$$

The term $\eta\pi_0(t,y)$ is the posterior expected cost of admitting a null-alpha signal.

False-discovery penalty

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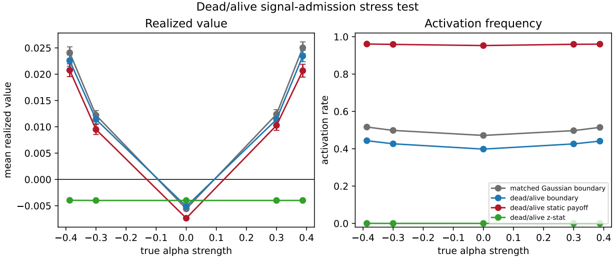
The term $\eta\pi_0(t,y)$ is the posterior expected cost of admitting a null-alpha signal.

$$\max\{\Phi_\eta - V,\, V_t + \mathcal{L}_t V - c\} = 0, \quad \mathcal{L}_t f = m(t,y) f_y + \tfrac{1}{2} f_{yy}.$$

With three states the posterior mean is no longer a sufficient coordinate, so the value is solved in *evidence space* y. There $\mathcal{L}_t$ is the generator of the posterior-predictive evidence $dY_t = m(t,Y_t)\,dt + d\widetilde{W}_t$; solved by a monotone upwind scheme.

Controlled validation of the three-state rule



Null activation

$$47.1\% \to 39.8\%$$

matched Gaussian boundary versus three-state boundary.

Null realized value

$$-0.00560 \to -0.00513$$

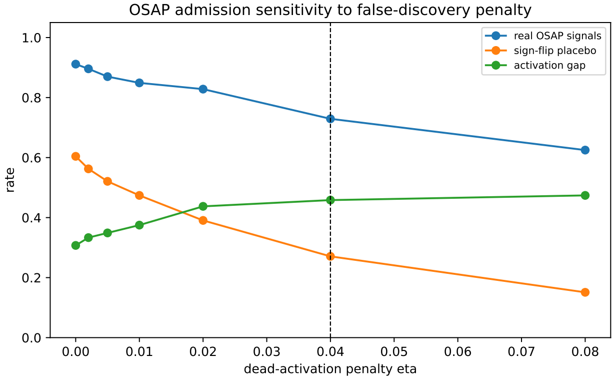
The rule trades lower null activation against slightly lower live-alpha value.

Real-data design

Data	Chen-Zimmermann Open-Source Asset Pricing (OSAP) long-sh
Eligible signals	192 predictors with enough history
Calibration	120 months
Sequential validation	up to 240 months
Holdout	120 post-decision months
Evidence increment	$\Delta Y_t = (r_t/\widehat{\sigma}_m)\sqrt{dt}$
Placebo	sign-flipped validation returns

Under a null with monthly variance $\widehat{\sigma}_m^2$, $\mathrm{Var}(\Delta Y_t) \approx dt$, so evidence is on Brownian scale.

Penalty calibration with real signals and placebos



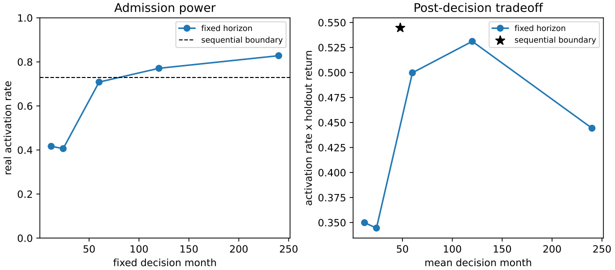
At $\eta = 0.04$

real activation $= 72.9\%$

placebo activation $= 27.1\%$

The placebo activation rate is more sensitive to the penalty than the real-signal activation rate.

Is the rule really sequential?



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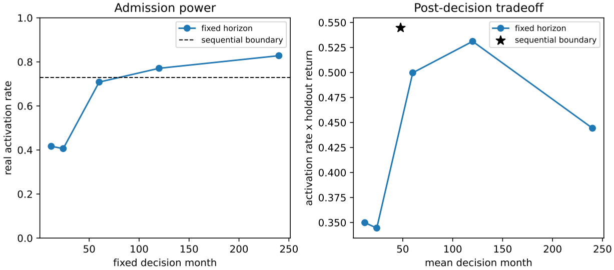


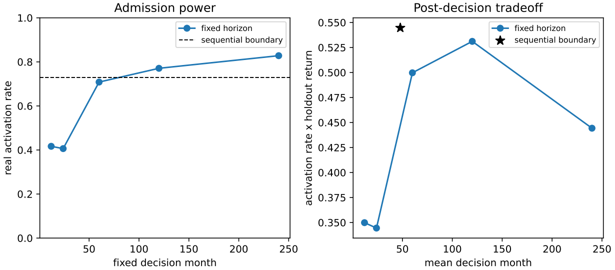
Sequential rule

72.9% real activation

47.9 month average decision time

0.545 activation-weighted holdout

Is the rule really sequential?



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47.9 month average decision time

0.545 activation-weighted holdout

Longer fixed horizons increase admission power, but at substantially later decision dates and lower activation-weighted holdout returns.

WRDS stock-level reconstruction

Return stream	N	Active	Standard active	Std. holdout
Gross stock-level	3	3	3	1.227%
Net range-cost stress	3	3	0	n/a

Three signals rebuilt from CRSP/Compustat via WRDS (Wharton Research Data Services). Gross long-short returns are admitted in the standard direction. The range-cost proxy (turnover $\times$ monthly high-low range) removes several percent per month, turning posterior net drift negative: the symmetric rule would then activate by *shorting*, so standard-direction admissions are zero.

The experiment separates statistical signal evidence from implementable signal value.

Mathematical and numerical checks

- Posterior probabilities remain positive and sum to one.
- $\partial_y G(t, y) = H(t, y)$ and $\partial_y m(t, y) = q(t, y)$ are checked numerically.
- Gaussian posterior precision update matches the closed form.
- The obstacle solver satisfies terminal value, symmetry, and monotonicity tests.
- Dead/alive finite-difference diagnostics: zero obstacle violation, zero terminal error, symmetry error 4.2×10^{-16}.
- Grid convergence: the paper grid has $V(0, 0)$ error below 9×10^{-5} relative to the fine grid.

What is novel here?

- The EKV filtering state is used for alpha validation rather than drift estimation alone.
- The terminal payoff is economic: posterior alpha must clear risk, uncertainty, and costs.
- The three-state prior makes the null-alpha probability part of the state.
- The empirical test asks whether sequential stopping beats fixed-horizon posterior tests at matched placebo activation.

One-line conclusion

Alpha validation is a sequential admission problem, not a fixed-date significance test.

Final takeaways

1. EKV gives a rigorous Bayesian learning state.
2. Economic tradeability changes the stopping reward.
3. Gaussian beliefs reveal the learning option value but exclude a point-mass null.
4. A three-state prior reduces null activation in simulation and placebo tests.
5. Real data suggest earlier admission at comparable placebo-calibrated power.

References

Ekstrom, Karatzas, and Vaicenavicius.
Bayesian Sequential Least-Squares Estimation for the Drift of a Wiener Process.
Stochastic Processes and their Applications, 2022.

Chen and Zimmermann.
Open Source Cross-Sectional Asset Pricing.
Critical Finance Review, 2022.

Chen and Velikov.
Zeroing In on the Expected Returns of Anomalies.
Journal of Financial and Quantitative Analysis, 2023.